

Business Management System (BMS) - Master Product Requirements Document (PRD)

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1. Executive Summary & Product Vision

The Business Management System (BMS) is an enterprise-grade SaaS platform designed to streamline corporate governance, project execution, and expense tracking for mid-sized organizations. The system guarantees that operational activities are dynamically authorized, tracked through customized pipelines, and seamlessly unified with financial accounting registers.

By unifying corporate divisions, request pipelines, sequential project milestones, sortable expense ledgers, and executive analytics, the system provides absolute visibility into corporate cash flows and departmental responsibilities.

2. Information Architecture & Navigation

The application employs a persistent left-hand sidebar navigation hierarchy to separate governance, daily operations, and financial auditing:

- **GOVERNANCE:** Houses the **Decisions** and **Approvals** interfaces.
- **OPERATIONS:** Houses the **Departments (Directory)** and **Actions** components.
- **FINANCE:** Houses the **Expenses** transaction ledger.
- **STANDALONE UTILITIES:** **Reports** and **Settings** reside at the base of the navigation panel.

Role-Based Access Control (RBAC) Matrix

To enforce a strict division of duties, system operations are mapped to user roles via explicit system permission scopes:

- **Administrators:** Full tenancy configuration, custom project pipeline management, and system audit log generation.
- **Managers:** Authorized to review, approve, or reject operational and financial requests.
- **Team Members:** Authorized to log draft decisions, update assigned action milestones, and submit expense receipts.

3. Relational Database Schema & Data Dictionary

The relational database architecture is designed for multi-tenant scalability, granular security compliance, custom workflow structures, and idle hooks for future extensions.

Entity-Relationship Diagram (ERD)

COMPANIES ||--o{ DEPARTMENTS : "owns"
COMPANIES ||--o{ USERS : "employs"
DEPARTMENTS ||--o{ USERS : "contains"
DEPARTMENTS ||--o{ ACTIONS : "executes"
USERS ||--o{ REQUESTS : "submits"
USERS ||--o{ REQUESTS : "resolves"
USERS ||--o{ SYSTEM_AUDIT_LOGS : "triggers"
REQUESTS ||--o{ ACTIONS : "spawns"
COMPANIES ||--o{ PIPELINES : "configures"
PIPELINES ||--o{ PIPELINE_STAGES : "defines"
ACTIONS ||--o{ EXPENSES : "records cost into"

Data Dictionary

Table 1: COMPANIES

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the tenant company.
name	VARCHAR(100)	NOT NULL	Legal business name.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of record creation.

Table 2: DEPARTMENTS

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the department.
company_id	INT	FK -> COMPANIES.id, NOT NULL	Associated tenant company.
name	VARCHAR(100)	NOT NULL	Department name (for example, Engineering, Finance, Sales).
code	VARCHAR(5)	NOT NULL	Department code for dynamic icon generation.
avatar_color_hex	VARCHAR(7)	NOT NULL	Design hex code for visual segmentation.

Column Name	Data Type	Constraints	Description
owner_user_id	INT	FK -> USERS.id, NULLABLE	Identifies the department lead.
owner_role_title	VARCHAR(100)	NULLABLE	Business title of the lead.
description	TEXT	NOT NULL	Core operational mandate.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of record creation.

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Table 3: USERS

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the user.
company_id	INT	FK -> COMPANIES.id, NOT NULL	Associated tenant company.
department_id	INT	FK -> DEPARTMENTS.id, NULLABLE	Primary department assignment of the user.
first_name	VARCHAR(50)	NOT NULL	First name of the user.
last_name	VARCHAR(50)	NOT NULL	Last name of the user.
email	VARCHAR(100)	UNIQUE, NOT NULL	Unique professional email.
role	VARCHAR(20)	NOT NULL	System access level (Admin, Manager, Employee).
two_factor_enabled	BOOLEAN	DEFAULT FALSE	Security status flag.
session_timeout_minutes	INT	DEFAULT 30	Active duration before session expiration.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of record creation.

Table 4: USER_PERMISSIONS

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the permission set.
user_id	INT	FK -> USERS.id, NOT NULL	Associated user record.
can_approve_financial	BOOLEAN	DEFAULT FALSE	Financial authorization privilege.
can_approve_operational	BOOLEAN	DEFAULT FALSE	Operational authorization privilege.

Column Name	Data Type	Constraints	Description
can_create_actions	BOOLEAN	DEFAULT FALSE	Permission to launch actionable tracking.
updated_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of last update.

Table 5: REQUESTS (Derived from Decisions Log)

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the decision request.
company_id	INT	FK -> COMPANIES.id, NOT NULL	Scoped tenant identifier.
title	VARCHAR(255)	NOT NULL	Short title of the decision request.
description	TEXT	NOT NULL	Detailed scope of the request.
submitted_by_user_id	INT	FK -> USERS.id, NOT NULL	The user who submitted the request.
estimated_cost	DECIMAL(12, 2)	DEFAULT 0.00, NOT NULL	Budget projection.
justification	TEXT	NOT NULL	Core operational explanation for the request.
status	VARCHAR(20)	NOT NULL	Lifecycle state: Pending, Approved, Rejected.
resolved_by_user_id	INT	FK -> USERS.id, NULLABLE	The manager who resolved the decision.
rejection_reason	TEXT	NULLABLE	Populated only if the status is Rejected.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of request submission.
resolved_at	TIMESTAMP	NULLABLE	Date and time of request resolution.

Table 6: PIPELINES (Custom Admin-Managed Workflows)

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the workflow pipeline.
company_id	INT	FK -> COMPANIES.id, NOT NULL	Associated tenant company.
name	VARCHAR(100)	NOT NULL	Pipeline name (for

Column Name	Data Type	Constraints	Description
			example, "Software Procurement").
is_default	BOOLEAN	DEFAULT FALSE	Designates the fallback template.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Date and time of pipeline creation.

Table 7: PIPELINE_STAGES

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the pipeline stage.
pipeline_id	INT	FK -> PIPELINES.id, NOT NULL	Parent pipeline container.
stage_name	VARCHAR(50)	NOT NULL	Stage label (for example, "Under Review").
sequence_order	INT	NOT NULL	Sequence index ensuring correct step progression.

Table 8: ACTIONS

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the operational task.
request_id	INT	FK -> REQUESTS.id, NOT NULL	Source request origin.
pipeline_stage_id	INT	FK -> PIPELINE_STAGES.id, NOT NULL	Current active workflow stage.
title	VARCHAR(255)	NOT NULL	Task title or objective.
description	TEXT	NOT NULL	Detailed instruction steps.
assigned_to_user_id	INT	FK -> USERS.id, NOT NULL	The user assigned to complete the action.
estimated_cost	DECIMAL(12, 2)	NOT NULL	Projected financial ceiling.
actual_cost	DECIMAL(12, 2)	NULLABLE	Validated real-world cost.
notes	TEXT	NULLABLE	Progress annotations.
due_date	DATE	NULLABLE	Operational deadline.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Task log creation time.

Table 9: EXPENSES

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the expense transaction.
company_id	INT	FK -> COMPANIES.id, NOT NULL	Associated tenant company.
title	VARCHAR(255)	NOT NULL	Spend label.
amount	DECIMAL(12, 2)	NOT NULL	Expense cost.
date	DATE	NOT NULL	Date the cost was incurred.
category	VARCHAR(50)	NOT NULL	Financial category (for example, Hardware, Travel).
origin_type	VARCHAR(20)	NOT NULL	Origin status (Manual or Action).
action_id	INT	FK -> ACTIONS.id, NULLABLE	Associated execution origin.
submitted_by_user_id	INT	FK -> USERS.id, NOT NULL	User who logged the expense.
status	VARCHAR(20)	NOT NULL	Approval status (for example, Logged, Synced).
ocr_extracted_data_raw	JSON	DEFAULT NULL, NULLABLE	OCR Provision Hook (Idle for future implementation).
ocr_processing_status	VARCHAR(20)	DEFAULT 'idle', NOT NULL	OCR State Flag (Idle for future implementation).
external_accounting_sync_id	VARCHAR(100)	DEFAULT NULL, NULLABLE	Accounting Integration Hook (Idle for future implementation).
external_sync_status	VARCHAR(20)	DEFAULT 'not_configured', NOT NULL	Accounting Sync State Flag (Idle for future implementation).
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Upload registration date.

Table 10: SYSTEM_AUDIT_LOGS (Detailed Compliance Logs)

Column Name	Data Type	Constraints	Description
id	INT	PK, Auto-Increment	Unique identifier for the

Column Name	Data Type	Constraints	Description
			audit event.
company_id	INT	FK -> COMPANIES.id, NOT NULL	Associated tenant company.
actor_user_id	INT	FK -> USERS.id, NOT NULL	The user who initiated the event.
ip_address	VARCHAR(45)	NOT NULL	Client IP address.
action_performed	VARCHAR(100)	NOT NULL	Event classification (for example, "Pipeline Modified").
target_table	VARCHAR(50)	NOT NULL	The table modified by the event.
target_id	INT	NOT NULL	Primary key of the modified record.
previous_state	JSON	NULLABLE	Values of the record prior to modification.
new_state	JSON	NULLABLE	Values of the record after modification.
created_at	TIMESTAMP	DEFAULT CURRENT_TIMESTAMP	Timestamp of log generation.

4. Deep-Dive Functional Modules

Module A: Expenses Ledger & Analytics

- **Target Dashboard Metrics:** Displays four central calculation summaries:
 - *Total Expenses:* Comprehensive historical spend.
 - *This Month:* Financial records matching the current calendar month.
 - *From Actions:* Sum of expenses linked to an active action card.
 - *Manual:* Sum of expenses manually populated by team members.
- **Interactive Filters & Sorting:** Tabs split views based on origin_type ("All", "Manual", "From Actions") alongside category-based dropdown queries. Rows allow dynamic ascending or descending sorting on **Date**, **Amount**, and **Title**.
- **Action Visual Linking:** Manual records are presented with clean attributes, whereas Action-spawned records render a linked action badge underneath the title. This allows users to trace expenses directly to their root operational actions.

Module B: Custom & Linear Action Pipelines

- **Pipeline Management:** Administrators can configure customized pipeline templates containing ordered workflow stages. If no custom pipeline is specified, actions fall back to the default linear system steps: `\text{Pending Order}` $\rightarrow$ `\text{Ordered}` $\rightarrow$ `\text{In Transit}` $\rightarrow$ `\text{Delivered}` $\rightarrow$ `\text{Completed}`
- **Project Progress Reporting:** Operational leads can transition actions through pipeline

- stages, updating milestone metrics such as Status, Execution Dates, and Actual Cost.
- **Financial Variance Calculations:** Upon reaching the terminal stage of a pipeline, the system evaluates financial variances: $\text{Variance} = \text{Actual Cost} - \text{Estimated Cost}$
 - A negative variance indicates cost savings and is formatted in green text (for example, $-\$300.00$).
 - A positive variance indicates budget overruns and is formatted in red text (for example, $+\$150.00$).
 - **Automated Expense Synchronization:** Updating an action to its terminal stage triggers a backend transaction that generates a linked record in the EXPENSES table, pre-populating fields using the confirmed actual cost.

Module C: Approvals State Machine

- **The Queue Panel:** Lists active requests waiting for sign-off. Cards display metadata, estimated costs, and justification explanations.
- **Approval Executions:** Approving a request requires selecting the specific manager signing off on the decision to establish accountability. This transition updates the status to Approved and generates an ACTION card mapped to the initial stage of the selected pipeline.
- **Rejection Executions:** Rejecting a request requires the manager to submit a written explanation in the rejection modal. This is saved as rejection_reason and displayed in the **Decision History** logs.
- **Turnaround Metrics:** The average response KPI is computed using the turnaround formula: $T_{\text{turnaround}} = \text{resolved_at} - \text{created_at}$

Module D: Department Directory & Responsibility Matrix

- **Dynamic Directory Cards:** Displays active company divisions. Each card shows the avatar initials of the department, the lead owner, description, and the count of active actions running through pipelines.
- **Operational Workload Aggregations:** Active actions are calculated dynamically using database query filters:

```
SELECT COUNT(*) FROM actions a
JOIN pipeline_stages s ON a.pipeline_stage_id = s.id
WHERE a.department_id = :department_id
AND s.sequence_order < (SELECT MAX(sequence_order) FROM pipeline_stages
WHERE pipeline_id = s.pipeline_id);
```
- **Responsibility Matrix:** A grid compiled at the base of the interface indexes operational departments, mapping divisions directly to owners to prevent organizational overlap.

Module E: Admin Security & Comprehensive System Audit Logging

- **Administrative Security Panel:** Administrators manage standard tenant-wide security properties, including two-factor authentication requirements, session limiters, and password age parameters.

- **Admin Audit Log Console:** A dedicated administrative interface displays system audit logs. Administrators can filter events by Date, Actor, and Event Category, and download logs for compliance reviews.
- **Security Event Tracking:** To prevent log tampering, the system records logs on a dedicated backend channel, logging events such as privilege changes, configuration updates, and request overrides.

5. API Endpoint Specifications

POST /api/v1/admin/pipelines

- **Access Level:** Administrators.
- **Description:** Creates a custom workflow pipeline template.
- **Request Payload:**

```
{
  "name": "Hardware Procurement",
  "stages": [
    { "stage_name": "Quote Requested", "sequence_order": 1 },
    { "stage_name": "Purchase Order Dispatched", "sequence_order": 2 },
    { "stage_name": "Assembled & Imaged", "sequence_order": 3 },
    { "stage_name": "Assigned to Asset Inventory", "sequence_order": 4 }
  ]
}
```

- **Response (201 Created):**

```
{
  "message": "Custom pipeline created successfully.",
  "pipeline_id": 12,
  "stages_created": 4
}
```

PUT /api/v1/actions/{id}/progress

- **Access Level:** Assigned Operational Leads.
- **Description:** Updates progress milestones and processes financial logs upon completion.
- **Request Payload:**

```
{
  "pipeline_stage_id": 8,
  "actual_cost": 4200.00,
  "notes": "Infrastructure completed within budget parameters."
}
```

- **Response (200 OK):**

```
{
  "message": "Action progress successfully updated."
}
```

```

"action_id": 87,
"current_stage_id": 8,
"is_terminal_stage": true,
"financials": {
  "estimated_cost": 4500.00,
  "actual_cost": 4200.00,
  "variance": -300.00
},
"expense_record_generated": {
  "expense_id": 1542,
  "status": "Logged"
}
}

```

GET /api/v1/admin/audit-logs

- **Access Level:** Administrators.
- **Description:** Retrieves tenant logs for administrative review.
- **Response (200 OK):**

```

{
  "logs": [
    {
      "id": 4012,
      "actor_user": "Martins Samuel Brengshak",
      "ip_address": "192.168.1.52",
      "action_performed": "Pipeline Modified",
      "target_table": "pipelines",
      "target_id": 12,
      "created_at": "2026-07-15T20:30:11Z"
    }
  ]
}

```

6. Market-Readiness Strategic Positioning

The architecture of the platform is structured to support enterprise integration, automated scanning, and advanced workflows, ensuring market competitiveness.

1. AI-Powered Receipt Processing (OCR)

- **Strategic Classification:** *Great-to-Have* (Deferred for the initial release).
- **Implementation Strategy:** This feature is not enabled for active UI deployment. However, the database schema includes idle fields (`ocr_extracted_data_raw` and `ocr_processing_status` inside the `EXPENSES` table) to support seamless future API integration.

2. Seamless Accounting Software Integration

- **Strategic Classification:** *Future Capability* (Design Optimized, Core Feature Deferred).
- **Implementation Strategy:** The architecture simplifies future integrations with accounting tools like QuickBooks, Xero, and Sage. Database schemas include standard foreign-mapping identifiers (external_accounting_sync_id and external_sync_status) to ensure future sync tools do not require complex table modifications.

3. Comprehensive System Audit Logging

- **Strategic Classification:** *Core Security Requirement* (Fully Integrated).
- **Implementation Strategy:** To meet corporate security standards, the system tracks and saves administrative operations, permission changes, and financial adjustments in the SYSTEM_AUDIT_LOGS table. Administrators can review, filter, and export these logs directly from the Admin console.

4. Admin-Managed Custom Project Pipelines

- **Strategic Classification:** *Core Usability Advantage* (Fully Integrated).
- **Implementation Strategy:** To support varied departmental workflows, the hardcoded linear milestone model is replaced with dynamic workflows. Administrators can create, order, and assign custom project pipelines to match different procurement types and departmental tasks.

7. Footnotes (Suggested Operational Improvements)

The following capabilities are excluded from the current functional specifications and development roadmap. They are documented here as future product updates:

- **Automated Expense Safeguard Validation**
- **Dynamic Departmental Budget Threshold Rules**
- **Automated Slack or Teams Notification Integration**